$(\forall x P(x)) \wedge A$ is True $\Rightarrow \forall x(P(x) \wedge A)$ is True.

Let, $\forall x(P(x) \wedge A)$ is True. By defn. of universal quantifier, for all elements a in the domain, $P(a) \wedge A$ is True. By defn. of AND, $P(a)$ is True and A is True for all elements a in the domain. By defn. of universal quantifier, $(\forall x P(x))$ is True. By defn. of AND, $(\forall x P(x)) \wedge A$ is True.

$\boxed{\therefore \forall x(P(x) \wedge A) \text{ is True} \Rightarrow (\forall x P(x)) \wedge A \text{ is True}}$

$\boxed{\therefore \text{They are logically equivalent}}$ $\square$

b) $(\exists x P(x)) \wedge A \equiv \exists x(P(x) \wedge A)$

Let $(\exists x P(x)) \wedge A$ be True. By defn. of AND, $\exists x P(x)$ is True and A is True. By defn. of existential quantifier, there exists an element a in the domain such that $P(a)$ is True. By defn. of AND, $P(a) \wedge A$ is True. By defn. of existential quantifier, $\exists x(P(a) \wedge A)$ is True.

$(\exists x P(x)) \wedge A$ is True $\Rightarrow \exists x(P(x) \wedge A)$ is True

Let, $\exists x(P(x) \wedge A)$ be True. By defn. of existential quantifier, there exists an element a in the domain s.t. $P(a) \wedge A$ is True. By defn. of AND, $P(a)$ is True and A is True. By defn. of existential quantifier, $\exists x(P(x))$ is True. By defn. of AND, $(\exists x P(x)) \wedge A$ is True.

$\exists x(P(x) \wedge A)$ is True $\Rightarrow (\exists x P(x)) \wedge A$ is True

$\boxed{\therefore \text{They are logically equivalent}}$ $\square$

50) Establish these logical equivalences, where x does not occur as a free variable in A. Assume that the domain is nonempty.

a) $\forall x(A \rightarrow P(x)) \equiv A \rightarrow \forall x P(x)$

Let, $\forall x(A \rightarrow P(x))$ be False. By defn. of negation, $\neg \forall x(A \rightarrow P(x))$ is True. By De Morgan's law of quantifiers, $\exists x \neg(A \rightarrow P(x))$ is True. By conditional disjunction equivalence, $\exists x \neg(\neg A \vee P(x))$ is True.